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Technologies

"A Good DIGITAL ENGINEER Must Be Adept In ANALOGUE ELECTRONICS To Be SUCCESSFUL"

How is technology shaping the future of electronics components? Rahul Chopra, editorial director, EFY Group, discusses the same in conversation with Anil Telikepalli, executive director of business management, industrial and healthcare, Maxim Integrated

Q. Is the IoT a buzz or reality?

A. A little bit of both, as devices are always being connected and automated! We keep hearing about smart things these days, and how these connected devices are multiplying. Whether we call it the Internet of Things (IoT) or something else, there is no denying that these intelligent things are here to stay.

We see the IoT as an opportunity for you to solve problems in new and better ways. We deliver technologies that help create innovative solutions. From a design perspective, smart and connected things commonly require sensing, processing, communication, power and security—all areas where Maxim can help.

It is typical to over-estimate the effect of a technology in the short run and underestimate its impact over the long term. Devices were always connected to one another in the past, too. Factories were always automated. What seems to have happened now is that this automation has started happening on a larger scale, and suddenly factories are beaming their data and reports to offices across the globe with easier access to Cloud computing.

Also, intelligence is being added into sensors and input/output that allows these to make decisions at their level without contacting the main controller. This enables higher throughput and adaptive manufacturing.

Q. Any specific product categories that are experiencing higher interests because of the Industrial IoT (or Manufacturing 4.0)?

A. We are seeing a broad demand for multiple product families at Maxim driven by Industry 4.0. Clearly, Maxim is a power powerhouse and with every system needing power supplies, we are seeing an insatiable demand for voltage regulators, system protection and supervisors that enable factory operators to reduce their expenses and downtime while improving output, resulting in higher profits.

Industry 4.0 also resulted in multiple new interfaces like IO-Link that have enabled adaptive manufacturing. In other words, operators can use the same factory assembly line to build multiple different products on demand, resulting in better factory utilisation.

Q. Energy management and efficiency is popular in markets such as India, so demand for products like Himalaya Series must be high. Any special initiatives that drive their demand among design engineers?

A. Yes! In fact, Maxim has a unique initiative in these regards. While we are a components' original equipment manufacturer (OEM), we also provide complete power modules for customers who want to go to market faster and focus their energies on solving system technical challenges other than power management.

In India this initiative has been successful, and we are able to work with a lot of local industrial-product OEMs. Himalaya is one of the series that has both components and power modules.

Q. More and more engineers are opting for the digital side of electronics. Experts cite shortage of good engineers with expertise in analogue. What is your view on this?

A. I think curiosity feeds innovation. My own experience includes digital and analogue electronics as well as software and fintech. A good digital engineer (namely, an RTL designer) must be adept in analogue electronics (understand voltages/currents through the circuit) to be successful. Similarly, a good analogue designer can become a star if she or he knows how to digitally process the acquired data.

Q. What are the top three technology trends that you believe will shape the future of electronic components?

A. I see three major trends. These are:

Artificial intelligence. New types of processors and algorithms are enabling artificial intelligence (AI) and machine learning that will have profound impact on electronic components, making these reconfigurable and adaptive.

Sensors. Nearly every segment, including industrial, healthcare, automotive and consumer, has realised that they can sense data and make decisions. Advancements in analogue technologies continue to drive this.

Electronics packaging. Semiconductor packaging is undergoing revolution to enable high integration. The day is not far when regular and even flexible PCBs will embed multiple die. EFY